Treatment: Eradication of amoebae by the use of amoebicidal drugs and replacement of fluid and electrolyte is the treatment for amoebiasis. Listed below the drugs used in the treatment for amoebiasis.

- Paramomycin and iodoquinol acts in the intestinal lumen but not in tissues.
- Emetine, chloroquine are effective in systemic infection. They act only on trophozoites.

Metronidazole is the drug of choice which acts as both luminal and tissue amoebicides. It is low in toxicity and is effective against intestinal as well as extra -intestinal amoebic infections.

8.2.8 Prevention and Control

- Proper sanitation is the key to avoid amoebiasis. Washing hands with soap and water after using the bathrooms and before handling food.
- Drinking safe and boiled water.
- Avoid eating unwashed fruits and vegetables.
- Prevention of water supplies from faecal contamination.
- Early rapid detection of diseased people and subsequent treatment with amoebicidal drugs. No vaccine is available yet against amoebiasis in humans.

8.3 Intestinal Flagellates – *Giardia Lamblia*

(Also known as *Giardia duodenalis, Giardia intestinalis*)

8.3.1 Geographical Distribution

It is the most common protozoan pathogen and is worldwide in distribution. The disease is very high in areas with low sanitation, especially tropics and subtropics.

8.3.2 Habitat

Giardia lamblia lives in the duodenum and upper jejunum of human. It is the only protozoan parasite found in the lumen of the human small intestine.

8.3.3 Morphology

It exists in two forms

- Trophozoite and
- Cyst



Antoine van Leewenhock observed and illustrated *Giardia lamblia* in his own loose stool. This was the first protozoan parasite of human that is recorded and the first to be seen under a microscope.

Trophozoite

The trophozoite is in the shape of a tennis or badminton racket. It is rounded anteriorly and pointed posteriorly. The size of the trophozoite is 14 µ long by 7µ broad. Dorsally, it is convex and ventrally, it has a concave sucking disc which helps in its attachment to the intestinal mucosa. It is bilaterally symmetrical. All the organs of the body are paired. Trophozoite of Giardia possess,

- 1 pair of nuclei
- 4 pairs of flagella
- Parabasal body (Blepharoplast), from which the flagella arise (4 pairs)

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